

## Assignment 2: Mastermind

### General

You complete this assignment alone or in pairs. If you work in pairs, each person must submit the assignment. At the top of your work, include a comment line stating with whom you collaborated.

This assignment counts for **18/30** of the daily work for the module: Learning to Program in Python, functional programming assignment.

You will be assessed on the correct functioning of the code, efficiency, and readability.

For this assignment you may use all course material provided in this module. You may also conduct online research to solve sub-problems. For example: how do I pick a random element from a list? How do I iterate alphabetically over a dictionary? etc. Do not look up complete solutions. You may not use AI (ChatGPT, Copilot, ...) to generate code for (parts of) the solutions.

Submit a Python file (.py) or a Notebook file (.ipynb) no later than **2 November** via Syntra Cloud.

### Context

Mastermind is a classic board game invented in the 1970s. There are two players: a codemaker and a codebreaker.

The codemaker creates a code consisting of a combination of 4 colored code pegs. They can choose from 6 different colors. The same color may be used more than once.

The codebreaker tries to guess the codemaker's code. They get at most 12 attempts. In each attempt the codebreaker makes a combination of 4 colored pegs. After each attempt, the codemaker provides feedback about the correctness of the guess. The codemaker places a black peg for each colored peg in the correct position. For each peg with a color that occurs in the code but is in the wrong position, a white peg is placed. The black and white pegs may be given in any order.

Example 1:    Code codemaker    ● ● ● ●  
                 Attempt codebreaker    ● ● ● ●    information pegs    ● ● ○

The red peg is correct; place a black peg.

The orange peg is incorrect; no peg is placed.

The green peg is correct but in the wrong position; place a white peg.

The yellow peg is correct; place a black peg.

Example 2:    Code codemaker    ● ● ● ●  
                 Attempt codebreaker    ● ● ● ●    information pegs    ● ○

The red peg is correct; place a black peg.

The orange peg is incorrect; no peg is placed.

The green peg is correct but in the wrong position; place a white peg.

The green peg is incorrect (because the previous green peg already counted as a misplaced peg); therefore no peg is placed.

### Assignment

Write a program for the Mastermind game where the user plays the codebreaker and the computer is the codemaker.

Input and output may be text-based. After each attempt by the user, show their guess together with feedback about its correctness.

The program should also indicate when the code has been cracked or when the 12 attempts are used up, in which case the computer wins.

For example G: yellow, B: blue, O: orange, R: red

CODE: G/B/O/O

INPUT: G/R/B/O

OUTPUT: G/R/B/O correct: 2, misplaced: 1

### Bonus 1 (2 pts)

Write the program so that the output displays colored dots for the guess and the feedback pegs.